- Cloud services like Google Cloud Storage and Azure Files continue to provide ACL-based models for fine-grained permissions on storage objects, even while also supporting more advanced IAM policies.

ACLs also work well in single-user or small-team applications, where adding a full role or policy engine would introduce more complexity than it's worth. In such cases, ACLs provide transparency, speed, and direct control.

Ultimately, ACLs are not obsolete; they're just specialized. When used in the appropriate context, they provide a solid foundation for access control, particularly in systems where the overhead of more dynamic models isn't justified.

5.1.5 When ACLs fall short

As organizations grow, their systems and associated access control needs become more complex. Managing access on a per-resource, per-principal basis doesn't scale well when there are hundreds or thousands of users and resources with nuanced access scenarios. Adding or removing access for even a single person can require updating multiple separate lists. More importantly, it becomes challenging to maintain visibility and control—especially when auditing access decisions is a regulatory, accreditation, or security requirement. Without a unified approach to manage and reason about access, ensuring compliance and demonstrating that the right people have the right access at the right time becomes nearly impossible.

More importantly, ACLs lack a method to express why someone has access. They record what someone can do with a specific resource, but they do not provide the business rationale for that access. For example, is Doug permitted to edit the campaign folder because he's on the marketing team, or due to a temporary collaboration? The ACL doesn't say—it only lists permissions.

These challenges create pressure to adopt access control models that accommodate organizational complexity while remaining manageable. One widely accepted approach is role-based access control (RBAC), which introduces the concept of roles—bundles of permissions tied to job functions